# Experiment 10

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**Subject Name: System Design** **Subject Code: 23CSH-314**

1. **Aim**: To design a **Food Delivery Platform (similar to Swiggy, Zomato)** that enables users to browse restaurants, place orders, track deliveries in real-time, and make secure payments with high availability and low latency.

2. **Objective**:

   - To design a scalable architecture for a food delivery system
   - To enable real-time order tracking using location services
   - To support order placement, cancellation, and status updates
   - To ensure reliability in order processing and payment handling
   - To implement delivery partner assignment and tracking
   - To optimize performance using caching and asynchronous processing.

3. **Tools Used:**
   - **Frontend (ReactJS, HTML, CSS, JavaScript)**
     - Built responsive UI for browsing restaurants and placing orders
     - Integrated APIs for dynamic menu and order tracking
   - **Backend (Node.js / Java / Spring Boot)**
     - Developed REST APIs for users, orders, delivery, and payments
     - Implemented business logic for order lifecycle
   - **Database (PostgreSQL / MySQL)**
     - Managed transactional data (orders, users, payments)
     - Ensured ACID properties for order consistency
   - **Redis**
     - Cached frequently accessed data (menus, restaurant list)
     - Reduced latency for read-heavy operations
   - **Message Queue (Kafka / RabbitMQ)**
     - Used for asynchronous processing (order → delivery assignment)
     - Improved scalability and fault tolerance
   - **Maps API (Google Maps / Mapbox)**
     - Used for real-time delivery tracking and ETA calculation
   - **Deployment (AWS / GCP, Docker, Load Balancer)**
     - Cloud-based scalable infrastructure
     - Containerized services for easy deployment

# 4. System Requirements:

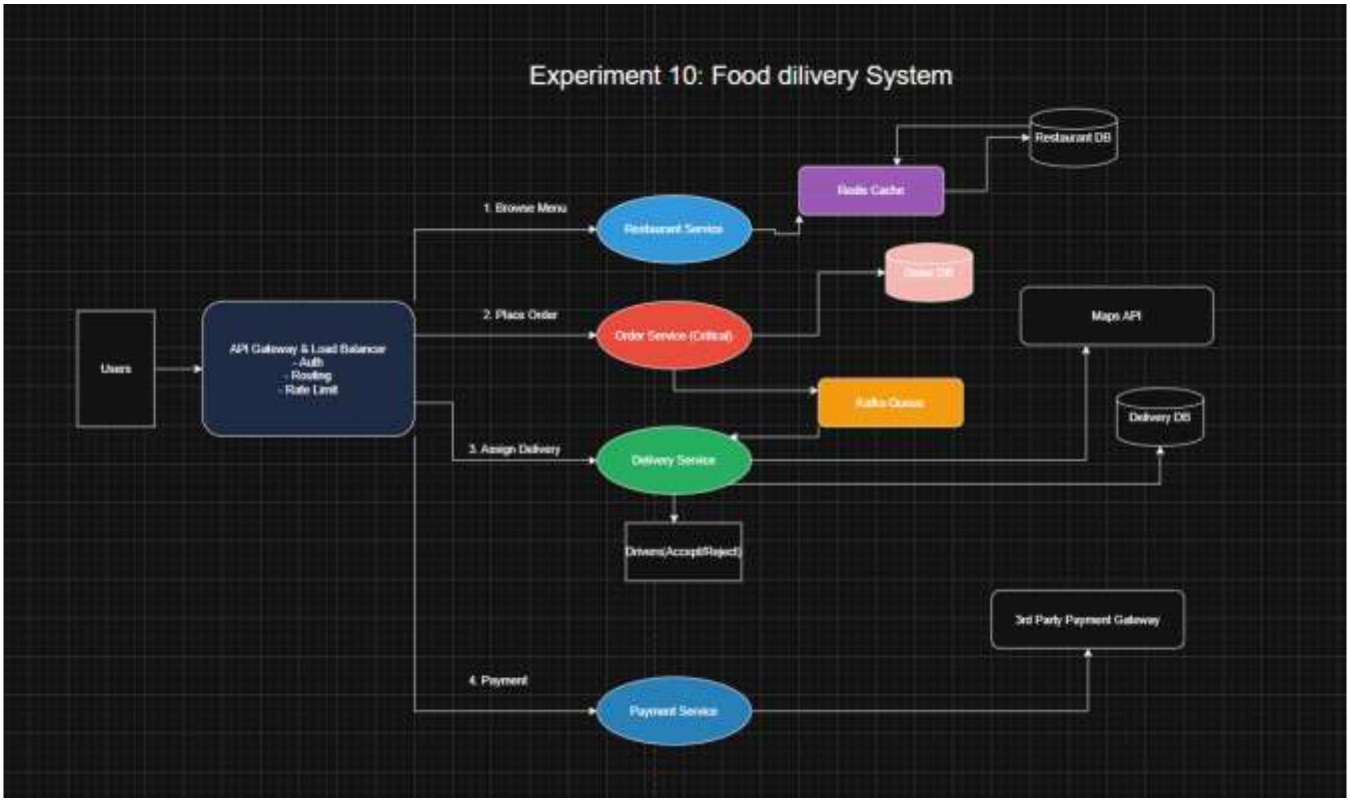
## A. Functional Requirements

- Users should be able to register and login
- Users should be able to browse restaurants and menus
- Users should be able to add items to cart and place orders
- The system should assign a delivery partner automatically
- Users should be able to track order status in real-time
- Users should be able to make payments (online / COD)
- Users should be able to cancel orders (before preparation stage)
- Restaurants should be able to accept/reject orders
- Delivery partners should be able to accept/decline delivery requests
- Users should be able to rate and review orders.

## B. Non-Functional Requirements

- **Scalability**
  System should support **millions of users and thousands of concurrent orders**, especially during peak hours. **CAP Theorem**
- **Consistency + Availability**
  Strong consistency → Order & Payment
  Eventual consistency → Tracking & notifications
- **Latency:**
  Order placement: **< 200 ms**
  Menu fetch: **< 100 ms**
  Tracking updates: **real-time (~1–2 sec delay)**
- **Reliability:**
  No duplicate orders
  Payment must be processed exactly once

## 5. HLD:




## 6. Scalability Solution

- Use **WebSockets** for real-time stock price updates
- Use **API Gateway + Load Balancer** to distribute traffic
  Use **Redis caching** for menu and restaurant data
  Use **Kafka (event-driven architecture)** for order processing
  Use **microservices architecture**:
- Order Service
- Delivery Service
- Payment Service
- Restaurant Service
- Use **database sharding** for large-scale data
  Use **horizontal scaling** for services
  Optimize read-heavy operations via caching
- Optimize read-heavy operations using caching

## 7. Learning Outcomes (What I Have Learnt)

- Understood architecture of large-scale food delivery systems
- Learned how to design microservices-based applications
- Understood importance of **asynchronous processing (Kafka)**
- Learned real-time tracking using location services
- Understood trade-offs between **consistency vs availability**
- Gained knowledge of scaling techniques (cache, queue, load balancing)
- Learned how real-world systems handle peak traffic efficiently